LAN96459F

Hardware Design Checklist (Managed Mode)

1.0 INTRODUCTION

This document provides a hardware design checklist for the Microchip LAN96459F on Managed mode. These checklist items should be followed when utilizing the LAN96459F in a new design. A summary of these items is provided in Section 13.0, "Hardware Checklist Summary". Detailed information on these subjects can be found in the corresponding sections:

- Section 2.0, "General Considerations"
- Section 3.0, "Power Supplies and Grounding"
- Section 4.0, "Pin Strapping"
- Section 5.0, "Reference Clocks"
- Section 6.0, "Port Configuration"
- Section 7.0, "SerDes Interface"
- Section 8.0, "RGMII/RMII"
- Section 9.0, "Management Interfaces"
- Section 10.0, "1588 Support (PTP)"
- Section 11.0, "JTAG Interface"
- Section 12.0, "Layout Considerations"

2.0 GENERAL CONSIDERATIONS

2.1 Required References

The LAN96459F designer should have the following document(s) on hand:

- *LAN9645xF Data Sheet*

2.2 Pin Check

- Check the pinout of the part against the data sheet. Ensure that all pins match the data sheet and are configured as intended input, output, or bidirectional. Do not rely on just the pin name of the bus interface between two connecting devices. The same pin name may be defined as an input or an output depending on the interface perspective. This is especially true for the RGMII and RMII pins as defined from the PHY perspective or MAC perspective.
- It is highly recommended to follow the reference design schematics as a basis and to retain the reference design's use of GPIO (parallel as well as serial) whenever possible to minimize software changes. Keep a log of major changes (for example, port numbering, PHY addresses, GPIO, and SGPIO), and provide this log when there is a design review or when starting software customization.

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3.0 POWER SUPPLIES AND GROUNDING

3.1 Power Supplies

The power and ground connections for the LAN96459F 128-Pin TQFP package are shown in Figure 3-1. The power and ground connections for the LAN96459F 156-Pin VQFN-DR package are shown in Figure 3-2. The power decoupling connections are shown in Figure 3-3.

FIGURE 3-1: 128-PIN TQFP POWER AND GROUND CONNECTIONS FOR LAN96459F

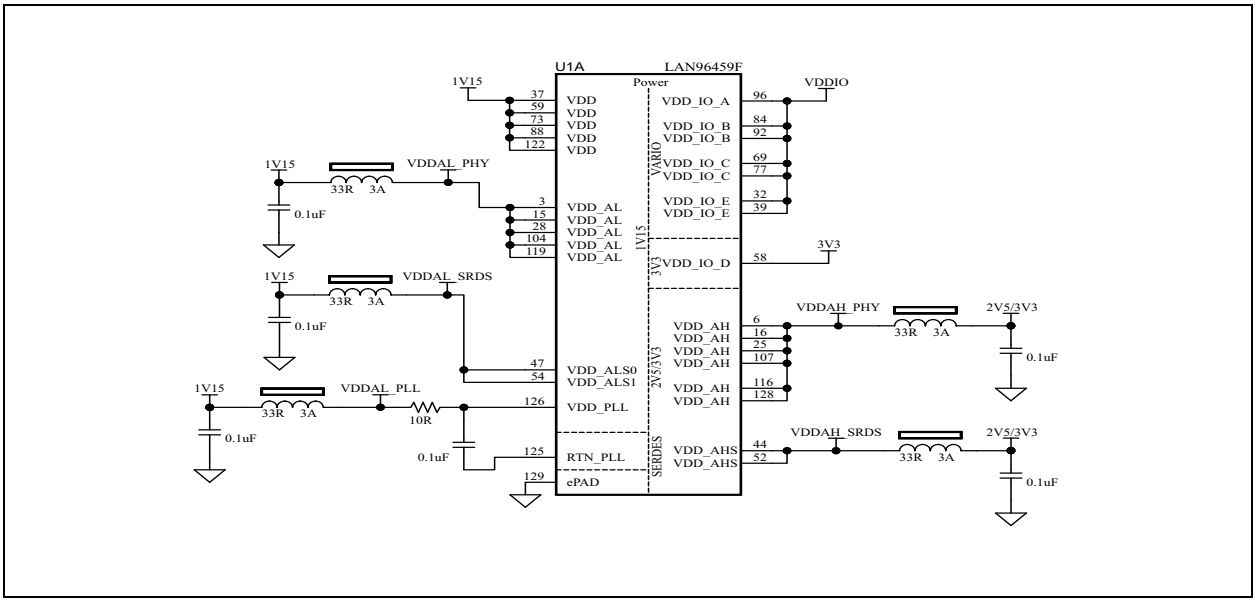
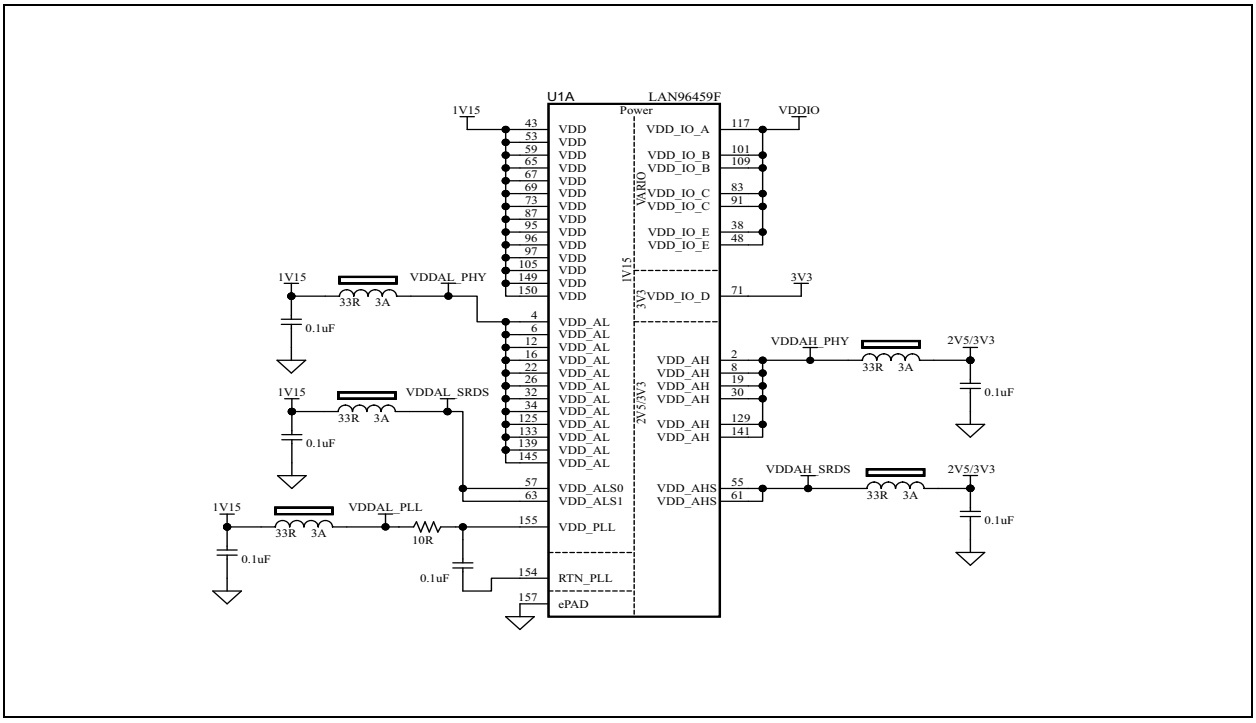
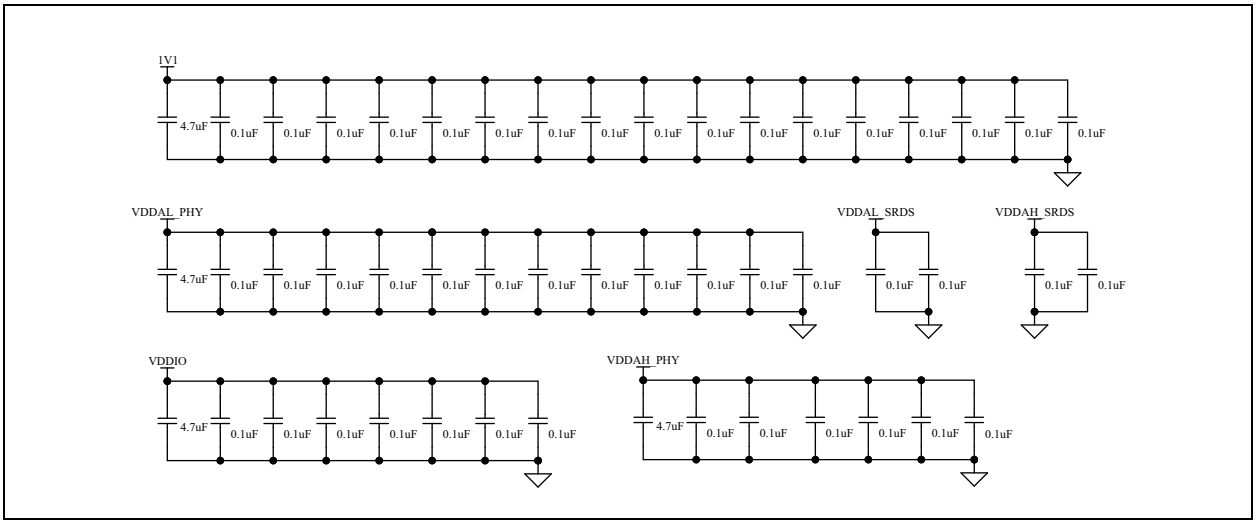


FIGURE 3-2: 156-PIN VQFN-DR POWER AND GROUND CONNECTIONS FOR LAN96459F



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FIGURE 3-3: POWER DECOUPLING CONNECTIONS



3.2 Power Pins

Table 3-1 shows the power supply pins for the LAN96459F.

TABLE 3-1: POWER SUPPLY PINS

Name	TQFP128 Pin	VQFN-DR156 Pin	Description
VDD	37, 59, 73, 88, 122	43, 53, 59, 65, 67, 69, 73, 87, 95, 96, 97, 105, 149, 150	1.15V Core power supply
VDD_AL	3, 15, 28, 104, 119	4, 6, 12, 16, 22, 26, 32, 34, 125, 133, 139, 145	1.15V Ethernet PHY analog low power supply
VDD_ALS0	47	57	1.15V SerDes 0 TX power supply
VDD_ALS1	54	63	1.15V SerDes 1 TX power supply
VDD_PLL	126	155	1.15V PLL power supply
VDD_IO_A	96	117	1.8/2.5/3.3V I/O Domain A power supply
VDD_IO_B	84, 92	101, 109	1.8/2.5/3.3V I/O Domain B power supply
VDD_IO_C	69, 77	83, 91	1.8/2.5/3.3V I/O Domain C power supply
VDD_IO_D	58	71	3.3V I/O Domain D power supply. (See **Note 1**.)
VDD_IO_E	32, 39	38, 48	1.8/2.5/3.3V I/O Domain E power supply
VDD_AH (PHY)	6, 16, 25, 107, 116, 128	2, 8, 19, 30, 129, 141	2.5/3.3V Ethernet PHY analog high power supply. (See **Note 2**.)
VDD_AHS (SERDES)	44, 52	55, 61	2.5/3.3V SerDes analog high power supply. (See **Note 2**.)

Note 1: Domain D power supply is fixed at 3.3V.

2: The supplies to VDD_AH and VDD_AHS must use the same voltage (2.5V or 3.3V).

3.3 Ground

Table 3-2 shows the ground pins for the LAN96459F.

TABLE 3-2: 128-PIN TQFP GROUND PINS

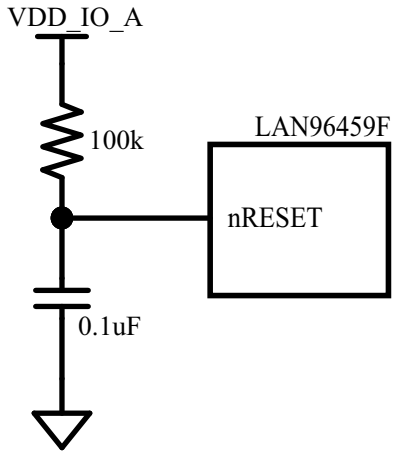
Name	TQFP128 Pin	VQFN-DR156 Pin	Description	Comments
RTN_PLL	125	154	PLL return	Do not connect to ground. Connect directly to the PLL decoupling capacitor.
ISET	127	156	Current calibration reference resistor connection pin for Ethernet PHYs	For proper operation, this pin must be connected to ground through a 6.0 kΩ 1% resistor.
ePAD	129	157	Ground	—

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3.4 RESET Circuit

The LAN96459F device is reset when the **nRESET** pin is low. The chip starts booting and becomes operational approximately 14 ms after **nRESET** has toggled high. The strapping pins are also latched when **nRESET** toggles high.

FIGURE 3-4: BASIC RESET CIRCUIT



Note 1: **nRESET** is pin 101 on the 128-Pin TQFP package, and pin 123 on the 156-Pin VQFN-DR package.

2: A Reset supervisor, such as MIC2774, may be used in place of the Reset circuit.

3: **nRESET** must be pulled up to the same voltage as **VDD_IO_A**.

4.0 PIN STRAPPING

4.1 Configuration Straps

Configuration straps allow various features of the device to be automatically configured to user-defined values. Configuration straps are latched upon the release of Reset pin (**nRESET**) and combined with any strap override register settings at chip top level. The results are then passed on to the functions using these straps. All strap signals have an internal pull-up. However, it is recommended to always connect an external pull-up/pull-down.

Note 1: The system designer must ensure that configuration strap pins meet timing requirements. In particular, the system designer must guarantee that the pins meet the timing requirements specified in the nRESET Timing Specifications section of the *LAN9645xF Data Sheet*. If the configuration strap pins are not at the correct voltage level prior to being latched, the device may capture incorrect strap values.

2: When externally pulling configuration straps high, the strap must be tied to the associated VDD_IO_x rail.

APPLICATION NOTE: All straps must be pulled up or pulled down externally on the PCB to enable the desired operational state.

Configuration strap definitions for the LAN96459F are specified in Table 4-1. For further details on the configuration straps, refer to the Configuration Straps section of the *LAN9645xF Data Sheet*.

TABLE 4-1: CONFIGURATION STRAPS

Strap Mode	Strap	GPIO	TQFP128 Pin	VQFN-DR156 Pin	Definition
SGPIO Enable	0	22	72	86	Used in GPIO mode A 0: Regular GPIOs used 1: SGPIOs used
SerDes 125 MHz	1	23	71	85	0: 125 MHz clock to SerDes disabled 1: 125 MHz clock to SerDes enabled (Only valid ID SGPIO_EN = 0)
Boot mode	4:2	26, 25, 24	67, 68, 70	81, 82, 84	000: No boot 001: I^2C boot – Reserved 010: SPI boot – Reserved 011: ROM – ROM boot 100: ROM + I^2C table 101: ROM + SPI table 110: ROM + VRAP port 5 111: ROM + VRAP port 7
Management Interface	6:5	28, 27	65, 66	79, 80	00: UART CLI 01: I^2C client 10: MIIM client 11: SPI client
JTAG	7	34	43	52	0: JTAG interface is enabled. 1: JTAG interface is disabled. If JTAG is enabled, pin strap #2 selects: 0: DFT TAP controller 1: SerDes TAP controller

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4.2 LED

TABLE 4-2: LED PIN NAMES

Name	Description	TQFP128 Pin-GPIO (Alt Assignment)	VQFN-DR156 Pin-GPIO (Alt Assignment)
PHY0_LED0	CuPHY LED output, two LEDs per PHY	Pin_33-GPIO_42 (ALT2), Pin_38-GPIO_38 (ALT3)	Pin_44-EGPIO_49 (ALT2), Pin_39-GPIO_42 (ALT2), Pin_47-GPIO_38 (ALT3)
PHY0_LED1	CuPHY LED output, two LEDs per PHY	Pin_31-GPIO_43 (ALT2)	Pin_45-EGPIO_50 (ALT2), Pin_36-GPIO_43 (ALT2)
PHY1_LED0	CuPHY LED output, two LEDs per PHY	Pin_35-GPIO_40 (ALT2), Pin_40-GPIO_37 (ALT3)	Pin_113-EGPIO_47 (ALT2), Pin_40-GPIO_40 (ALT2), Pin_49-GPIO_37 (ALT3)
PHY1_LED1	CuPHY LED output, two LEDs per PHY	Pin_34-GPIO_41 (ALT2)	Pin_46-EGPIO_48 (ALT2), Pin_41-GPIO_41 (ALT2)
PHY2_LED0	CuPHY LED output, two LEDs per PHY	Pin_38-GPIO_38 (ALT2), Pin_41-GPIO_36 (ALT3)	Pin_47-GPIO_38 (ALT2), Pin_50-GPIO_36 (ALT3)
PHY2_LED1	CuPHY LED output, two LEDs per PHY	Pin_36-GPIO_39 (ALT2)	Pin_114-EGPIO_46 (ALT2), Pin_42-GPIO_39 (ALT2)
PHY3_LED0	CuPHY LED output, two LEDs per PHY	Pin_41-GPIO_36 (ALT2), Pin_42-GPIO_35 (ALT3)	Pin_50-GPIO_36 (ALT2), Pin_51-GPIO_35 (ALT3)
PHY3_LED1	CuPHY LED output, two LEDs per PHY	Pin_40-GPIO_37 (ALT2)	Pin_49-GPIO_37 (ALT2)
PHY4_LED0	CuPHY LED output, two LEDs per PHY	Pin_43-GPIO_34 (ALT2), Pin_43-GPIO_34 (ALT3)	Pin_52-GPIO_34 (ALT2), Pin_52-GPIO_34 (ALT3)
PHY4_LED1	CuPHY LED output, two LEDs per PHY	Pin_42-GPIO_35 (ALT2)	Pin_51-GPIO_35 (ALT2)
MAC_LED0	LED output based on MAC status	Pin_42-GPIO_35 (ALT5)	Pin_51-GPIO_35 (ALT5)
MAC_LED1	LED output based on MAC status	Pin_41-GPIO_36 (ALT5)	Pin_50-GPIO_36 (ALT5)
MAC_LED2	LED output based on MAC status	Pin_40-GPIO_37 (ALT5)	Pin_49-GPIO_37 (ALT5)
MAC_LED3	LED output based on MAC status	Pin_38-GPIO_38 (ALT5)	Pin_47-GPIO_38 (ALT5)
MAC_LED4	LED output based on MAC status	Pin_36-GPIO_39 (ALT5)	Pin_42-GPIO_39 (ALT5)
MAC_LED5	LED output based on MAC status	Pin_35-GPIO_40 (ALT5)	Pin_40-GPIO_40 (ALT5)
MAC_LED6	LED output based on MAC status	Pin_34-GPIO_41 (ALT5)	Pin_41-GPIO_41 (ALT5)
MAC_LED7	LED output based on MAC status	Pin_33-GPIO_42 (ALT5)	Pin_39-GPIO_42 (ALT5)
MAC_LED8	LED output based on MAC status	Pin_31-GPIO_43 (ALT5)	Pin_36-GPIO_43 (ALT5)

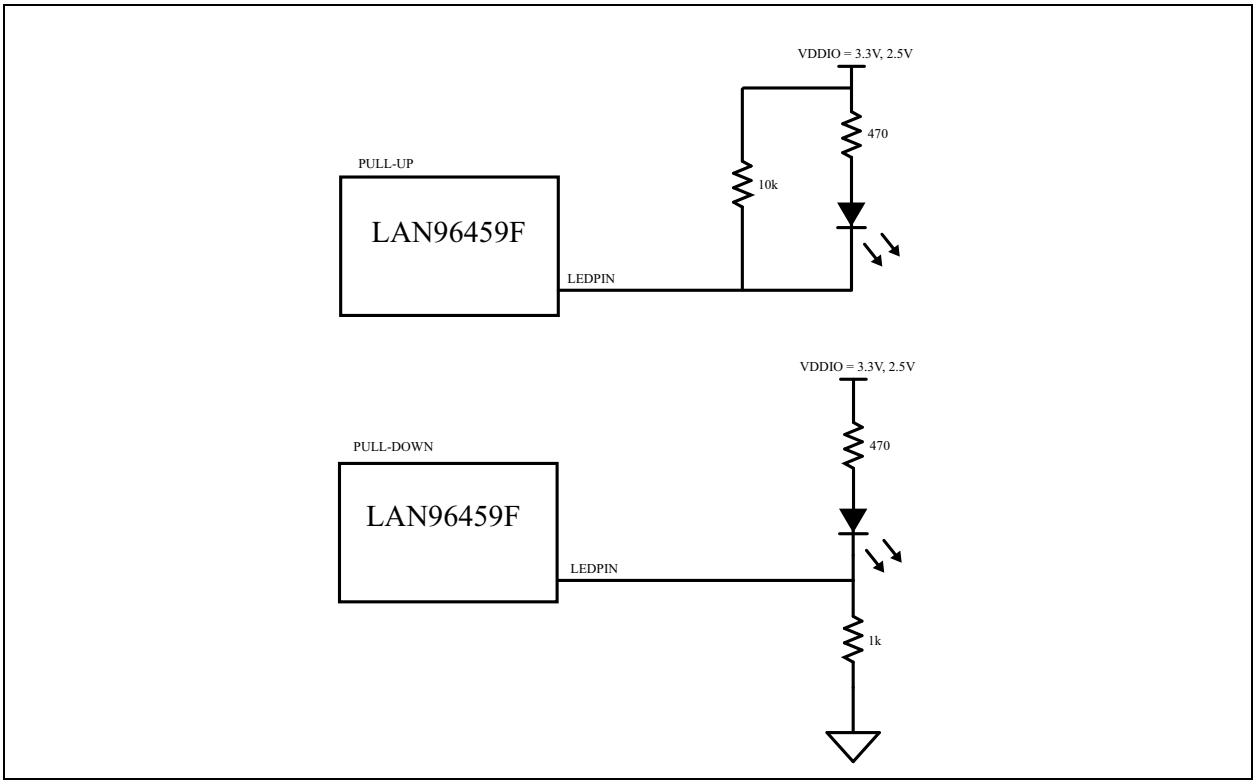
There is an overlap between LED pins and configuration straps. (Refer to Table 4-3.) To strap these pins while maintaining LED functionality, refer to Figure 4-1.

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TABLE 4-3: GPIO LED AND STRAP OVERLAP

GPIO	TQFP128 Pin	VQFN-DR156 Pin	Strap Function	LED Pin Name
34	43	52	JTAG	PHY4_LED0
35	42	51	Loop Detection	PHY4_LED1, MAC_LED0
36	41	50	GPIO mode	PHY3_LED0, MAC_LED1
37	40	49	GPIO mode	PHY3_LED1, MAC_LED2
38	38	47	RGMII1 mode	PHY2_LED0, MAC_LED3
39	36	42	SD0 mode	PHY2_LED1, MAC_LED4
40	35	40	SD1 mode	PHY1_LED0, MAC_LED5
41	34	41	RGMII1 mode	PHY1_LED1, MAC_LED6

FIGURE 4-1: LED STRAP CONFIGURATION DIAGRAMS



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5.0 REFERENCE CLOCKS

5.1 Layer 1 Timing

There are two recovered clock outputs that provide timing sources for external timing circuitry in redundant timing implementations.

The recovered clock outputs are mapped on GPIO overlaid functions. For more information, refer to the Section 3.3 GPIO Alternate Functions of the *LAN9645xF Data Sheet*.

It is possible to recover receive timing from any of the Ethernet ports. For aggregated port interfaces like QSGMII links, timing must be recovered in the PHY because the QSGMII link retimes during aggregation of data streams.

The HSIO:SYNC_ETH_CFG register provides per recovered clock divider configuration, allowing division before sending out clock frequency.

The SerDes recovered clock outputs also have individual divider configuration (through HSIO:SD[0-1]:SYNC_ETH_SD_CFG) to allow division of receive frequency. Note that division with 66/32 results in a gapped clock that is typically not recommended for SyncE operation.

The recovered clocks are single-ended outputs. The suggested divider settings in Table 5-1 to Table 5-4 are selected to avoid producing extremely high frequency via the input and outputs. A maximum frequency of 62.5 MHz is allowed.

TABLE 5-1: SERDES RECOVERED CLOCK SETTINGS FOR 1 GE

Output Frequency	Register Settings for Output n, Land m
25 MHz	SYNC_ETH_CFG[n].RECO_CLK_ENA=1, SYNC_ETH_CFG[n].SEL_RECO_CLK_DIV=4, and SYNC_ETH_CFG[n].SEL_RECO_CLK_SRC=(m) SD[m]::SYNC_ETH_SD_CFG.SD_RECO_CLK_DIV=0
31.25 MHz	SYNC_ETH_CFG[n].RECO_CLK_ENA=1, SYNC_ETH_CFG[n].SEL_RECO_CLK_DIV=1, and SYNC_ETH_CFG[n].SEL_RECO_CLK_SRC=(m) SD[m]:SYNC_ETH_SD_CFG.SD_RECO_CLK_DIV=0

Note 1: There are internal dividers for each interface and data rate.

TABLE 5-2: SERDES RECOVERED CLOCK SETTINGS FOR 2.5 GE

Output Frequency	Register Settings for Output n, Land m
31.25 MHz	SYNC_ETH_CFG[n].RECO_CLK_ENA=1, SYNC_ETH_CFG[n].SEL_RECO_CLK_DIV=4, and SYNC_ETH_CFG[n].SEL_RECO_CLK_SRC=(m) SD[m]:SYNC_ETH_SD_CFG.SD_RECO_CLK_DIV=1
62.50 MHz	SYNC_ETH_CFG[n].RECO_CLK_ENA=1, SYNC_ETH_CFG[n].SEL_RECO_CLK_DIV=4, and SYNC_ETH_CFG[n].SEL_RECO_CLK_SRC=(m) SD[m]:SYNC_ETH_SD_CFG.SD_RECO_CLK_DIV=0

TABLE 5-3: SERDES RECOVERED CLOCK SETTINGS FOR 1000BASE-T

Output Frequency	Register Settings for Output n, Land m
31.25 MHz	SYNC_ETH_CFG[n].RECO_CLK_ENA=1, SYNC_ETH_CFG[n].SEL_RECO_CLK_DIV=1, and SYNC_ETH_CFG[n].SEL_RECO_CLK_SRC=(m+4)
62.50 MHz	SYNC_ETH_CFG[n].RECO_CLK_ENA=1, SYNC_ETH_CFG[n].SEL_RECO_CLK_DIV=0, and SYNC_ETH_CFG[n].SEL_RECO_CLK_SRC=(m+4)

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TABLE 5-4: SERDES RECOVERED CLOCK SETTINGS FOR 1000BASE-T

Output Frequency	Register Settings for Output n, Land m
25 MHz	SYNC_ETH_CFG[n].RECO_CLK_ENA=1, SYNC_ETH_CFG[n].SEL_RECO_CLK_DIV=6, and SYNC_ETH_CFG[n].SEL_RECO_CLK_SRC=(m+4)
5 MHz	SYNC_ETH_CFG[n].RECO_CLK_ENA=1, SYNC_ETH_CFG[n].SEL_RECO_CLK_DIV=4, and SYNC_ETH_CFG[n].SEL_RECO_CLK_SRC=(m+4)

It is possible to automatically squelch the clock output when the devices detect a loss of signal or PCS loss of lock on an incoming data stream. This can be used for failover in external timing recovery solutions. Auto squelching is enabled by setting SD[n]:SYNC_ETH_SD_CFG.SD_AUTO_SQUELCH_ENA for the SerDes ports and CHIP TOP:CuPHY CFG(n):CuPHY PORT CFG.AUTO SQUELCH ENA for the CuPHY ports. To enable auto squelch on the CuPHY port, it is necessary to enable the Fast Link Fail mode first. This is done by setting the “FLF Enable” and “Enable FLF Link Down” bits in PHY register EP 0.15 (bit 1 and 0).

The SerDes signal detect pin can also be used to squelch the recovered clock. This is enabled by setting SD[n]:SIGDET_CFG.SD_ENA and should be used when the ports are connected to optical modules.

When squelching the clock, the clock will stop when detecting loss of signal (or PCS lock) either on high or low level.

5.2 Clocking

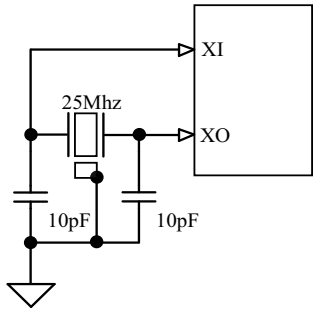
The following reference clocks can be provided to the device:

- Single-ended 25 MHz reference on **XI** pin: Can be generated using a clock crystal (XTAL) connected between the **XI** and **XO** pins or by using an oscillator connected to the **XI** pin. This clock is mandatory.
- 125 MHz differential clock on **S0_CLKP** pin: The 125 MHz Clock mode is enabled by pin strapping. When SerDes is used in QSGMII mode, SerDes can be configured to use the 125 MHz clock. If the 25 MHz mode is selected for both CuPHY and SerDes, the 125 MHz clock is not used and must not be connected.

The switch core and CPU clock generator PLLs use an internal 25 MHz clock as reference. When the external reference for CuPHY is strapped for 125 MHz operation mode, the internal 25 MHz PLL reference is generated by dividing the 125 MHz clock by 5. The internal 25 MHz clock can also be supplied using the crystal (XTAL) connected to the **XI** and **XO** pins or by using an oscillator connected to the **XI** pin. The CuPHY and SerDes can use either 25 MHz or 125 MHz reference clock. However, if any of the SerDes are running in QSGMII mode, it must be configured to use the 125 MHz reference clock. Refer to the SerDes 125 MHz Selector (SD_125M) of the *LAN9645xF Data Sheet* for details on how this is selected.

5.2.1 CRYSTAL CIRCUIT

FIGURE 5-1: CRYSTAL CIRCUIT



Note 1: **XI** is pin 124 on the 128-Pin TQFP package, and pin 153 on the 156-Pin VQFN-DR package.

2: **XO** is pin 123 on the 128-Pin TQFP package, and pin 152 on the 156-Pin VQFN-DR package.

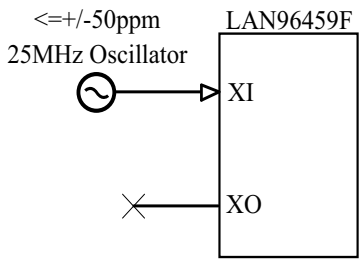
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5.2.2 EXTERNAL CLOCK SOURCE/OSCILLATOR CIRCUIT

If the single-ended 25 MHz reference clock is connected to an external clock source/oscillator, see Figure 5-2 for reference.

If the 125 MHz reference clock input is used, see Figure 5-3 for reference. Note that the resistor values vary depending on the oscillator output value.

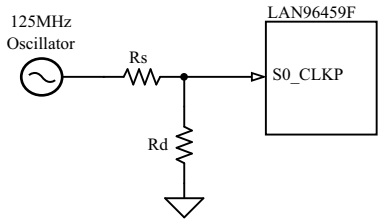
FIGURE 5-2: EXTERNAL25 MHZ CLOCK SOURCE/OSCILLATOR CIRCUIT



Note 1: **XI** is pin 124 on the 128-Pin TQFP package, and pin 153 on the 156-Pin VQFN-DR package.

2: **XO** is pin 123 on the 128-Pin TQFP package, and pin 152 on the 156-Pin VQFN-DR package.

FIGURE 5-3: EXTERNAL 125 MHZ CLOCK SOURCE/OSCILLATOR CIRCUIT



Note 1: **S0_CLKP** is pin 45 on the 128-Pin TQFP package, and pin 54 on the 156-Pin VQFN-DR package.

5.3 CK25OUT

The 156-Pin VQFN-DR package can also output a 25 MHz clock from PLL, which is present on pin 151. This can be used as a reference clock for external PHYs or recovered clock going back to a host processor.

If disabled by writing to the CK25OUT_ENA bit of the CUPHY_COMMON_CFG register, this pin should be left unconnected to prevent EMI.

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6.0 PORT CONFIGURATION

6.1 Logical Ports and Ethernet Interfaces

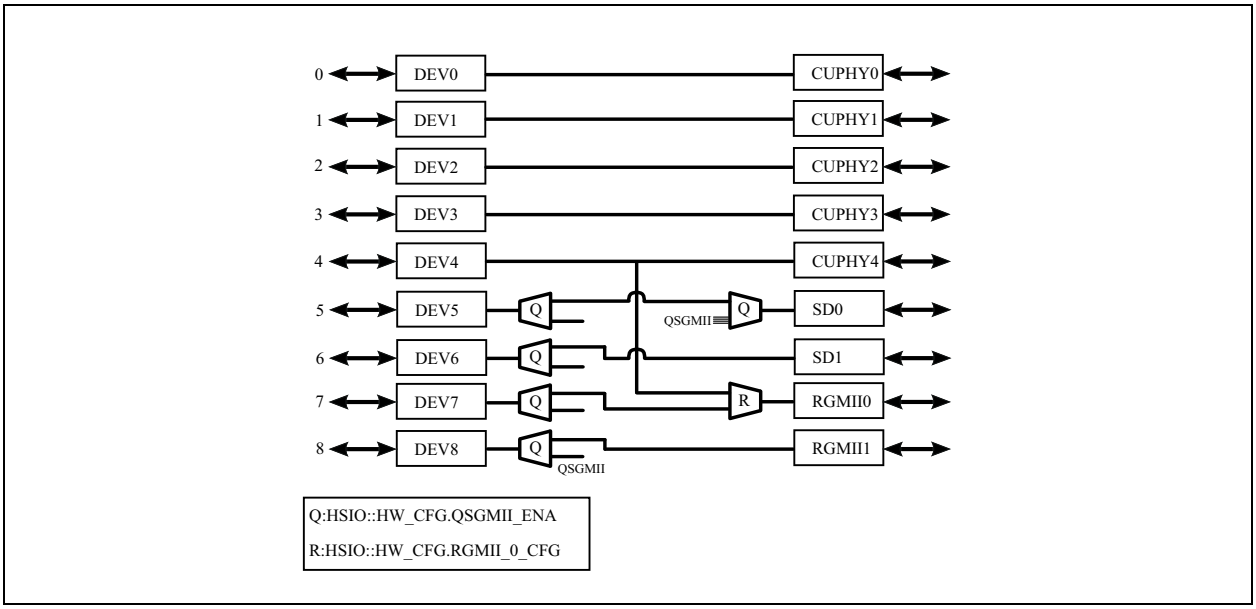
The device provides five (5), seven (7), or nine (9) Ethernet ports and one (1) internal CPU port. The device has a total of nine (9) Ethernet interfaces consisting of five (5) copper transceivers, two (2) SerDes, and two (2) RGMII interfaces. One of the SerDes supports QSGMII.

The switch core contains an internal port numbering domain where ports are numbered from 0 through 9. A port connects to a port module, and the module connects to an interface macro. The macro connects to I/O pins on the device. When using RGMII, the port module connects directly to the RGMII pins without the need for an interface macro.

6.1.1 PORT MULTIPLEXING

Port multiplexing defines how the logical Ethernet ports connect to the Ethernet interfaces. Port multiplexing is controlled through register HSIO::HW_CFG. Figure 6-1 highlights the available port multiplexing.

FIGURE 6-1: PORT MULTIPLEXING



As a result of port multiplexing, the device supports a range of port configurations. Table 6-1 lists the active ports in major supported port configurations. Table 6-2 details the port numbers as they correspond with modules, speeds, and Ethernet interface options.

TABLE 6-1: PORT CONFIGURATIONS

Port Configuration	CuPHY #0	CuPHY #1	CuPHY #2	CuPHY #3	CuPHY #4	SGMII #0	SGMII # 1	RGMII #0	RGMII #1
5x CuPHY + 2x 2.5G SGMII + 2x RGMII	DEV0	DEV1	DEV2	DEV3	DEV4	DEV5	DEV6	DEV7	DEV8
5x CuPHY + 1x QSGMII	DEV0	DEV1	DEV2	DEV3	DEV4	DEV5 DEV6 DEV7 DEV8	—	—	—
4x CuPHY + 1x QSGMII + 1x RGMII	DEV0	DEV1	DEV2	DEV3	—	DEV5 DEV6 DEV7 DEV8	—	DEV4	—

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TABLE 6-2: PORT NUMBERING

Port Number	Port Module	Maximum Port Speed	Ethernet Interface Options
0	DEV0	1 Gbps	CuPHY #0
1	DEV1	1 Gbps	CuPHY #
2	DEV2	1 Gbps	CuPHY #
3	DEV3	1 Gbps	CuPHY #
4	DEV4	1 Gbps	CuPHY # or RGMII #0
5	DEV5	2.5 Gbps	SerDes #0
6	DEV6	2.5 Gbps	SerDes #1 or SerDes #0 (QSGMII)
7	DEV7	1 Gbps	RGMII #0 or SerDes #0 (QSGMII)
8	DEV8	1 Gbps	RGMII #1 or SerDes #0 (QSGMII)
9	No port module	—	Internal (**Note 1**)

Note 1: Port 9 is an internal port used for manual injection and extraction of frames towards a CPU or processing of VRAP frames. It does not have an associated port module.

6.2 CuPHY Interfaces

The LAN96459F has five (5) CuPHY ports from PHY 0 to PHY 4 for port 0, port 1, port 2, port 3, and port 4. All these ports are AIO buffered. Therefore, coupling capacitors and differential routing are required (impedance value = 100) and without external termination. Detailed pin numbers from PHY 0 to PHY 4 sequence and descriptions are shown in Table 6-3.

TABLE 6-3: CUPHY INTERFACE PINS

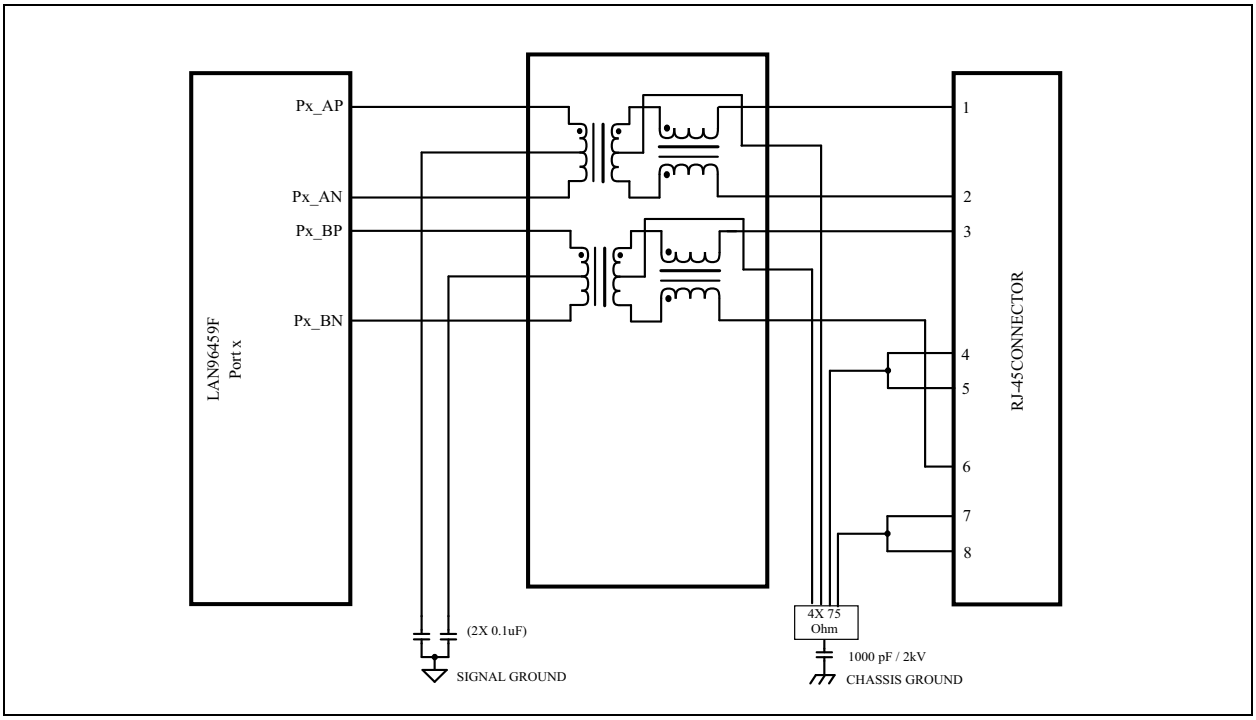
PHY	Name	TQFP128 Pin	VQFN-DR156 Pin	Description
PHY0	P0_AP	102	122	10/100/1000 Port 0, Channel A
PHY0	P0_AN	103	124	10/100/1000 Port 0, Channel A
PHY0	P0_BP	105	126	10/100/1000 Port 0, Channel B
PHY0	P0_BN	106	128	10/100/1000 Port 0, Channel B
PHY0	P0_CP	108	130	10/100/1000 Port 0, Channel C
PHY0	P0_CN	109	132	10/100/1000 Port 0, Channel C
PHY0	P0_DP	110	134	10/100/1000 Port 0, Channel D
PHY0	P0_DN	111	135	10/100/1000 Port 0, Channel D
PHY1	P1_AP	112	136	10/100/1000 Port 1, Channel A
PHY1	P1_AN	113	138	10/100/1000 Port 1, Channel A
PHY1	P1_BP	114	140	10/100/1000 Port 1, Channel B
PHY1	P1_BN	115	142	10/100/1000 Port 1, Channel B
PHY1	P1_CP	117	143	10/100/1000 Port 1, Channel C
PHY1	P1_CN	118	144	10/100/1000 Port 1, Channel C
PHY1	P1_DP	120	146	10/100/1000 Port 1, Channel D
PHY1	P1_DN	121	148	10/100/1000 Port 1, Channel D
PHY2	P2_AP	1	1	10/100/1000 Port 2, Channel A
PHY2	P2_AN	2	3	10/100/1000 Port 2, Channel A
PHY2	P2_BP	4	5	10/100/1000 Port 2, Channel B
PHY2	P2_BN	5	7	10/100/1000 Port 2, Channel B
PHY2	P2_CP	7	9	10/100/1000 Port 2, Channel C

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TABLE 6-3: CUPHY INTERFACE PINS (CONTINUED)

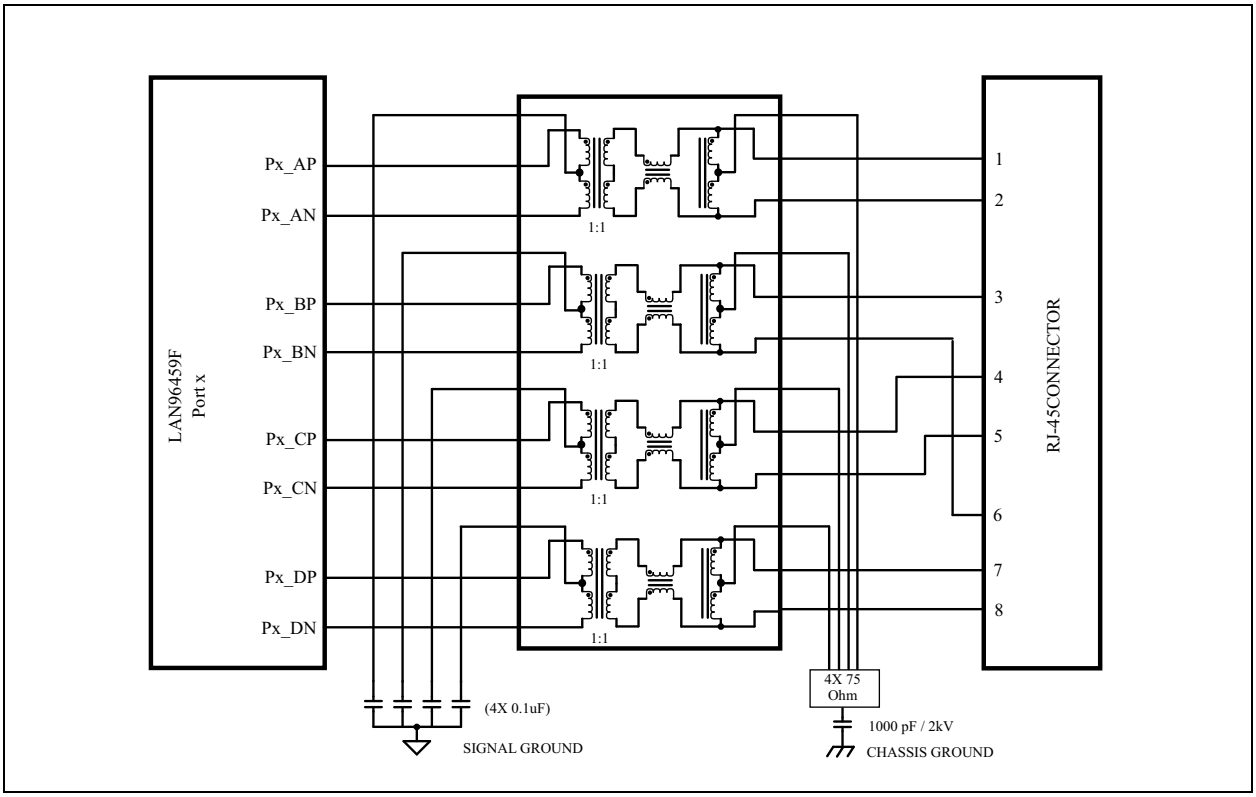
PHY	Name	TQFP128 Pin	VQFN-DR156 Pin	Description
PHY2	P2_CN	8	10	10/100/1000 Port 2, Channel C
PHY2	P2_DP	9	11	10/100/1000 Port 2, Channel D
PHY2	P2_DN	10	13	10/100/1000 Port 2, Channel D
PHY3	P3_AP	11	14	10/100/1000 Port 3, Channel A
PHY3	P3_AN	12	15	10/100/1000 Port 3, Channel A
PHY3	P3_BP	13	17	10/100/1000 Port 3, Channel B
PHY3	P3_BN	14	18	10/100/1000 Port 3, Channel B
PHY3	P3_CP	17	20	10/100/1000 Port 3, Channel C
PHY3	P3_CN	18	21	10/100/1000 Port 3, Channel C
PHY3	P3_DP	19	23	10/100/1000 Port 3, Channel D
PHY3	P3_DN	20	24	10/100/1000 Port 3, Channel D
PHY4	P4_AP	21	25	10/100/1000 Port 4, Channel A
PHY4	P4_AN	22	27	10/100/1000 Port 4, Channel A
PHY4	P4_BP	23	28	10/100/1000 Port 4, Channel B
PHY4	P4_BN	24	29	10/100/1000 Port 4, Channel B
PHY4	P4_CP	26	31	10/100/1000 Port 4, Channel C
PHY4	P4_CN	27	33	10/100/1000 Port 4, Channel C
PHY4	P4_DP	29	35	10/100/1000 Port 4, Channel D
PHY4	P4_DN	30	37	10/100/1000 Port 4, Channel D

FIGURE 6-2: 10/100 MB COPPER PHY CONNECTIONS



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FIGURE 6-3: 1000MB COPPER PHY CONNECTIONS



7.0 SERDES INTERFACE

7.1 Serial Data Pins

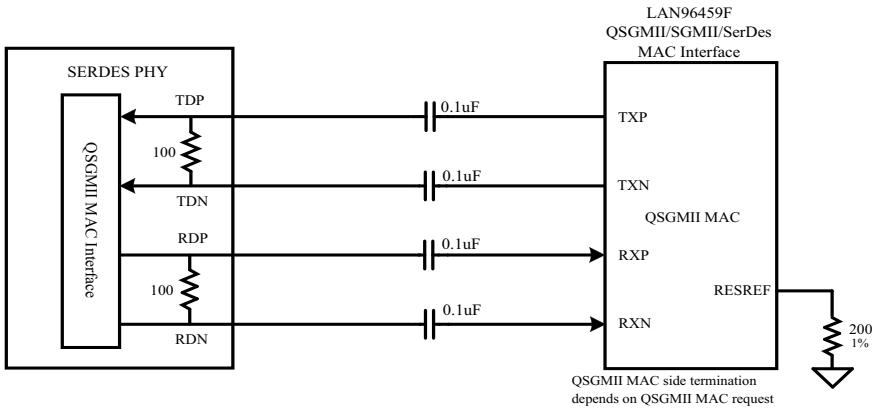
TABLE 7-1: SERIAL DATA PINS

Pin Name	TQFP128 Pin	VQFN-DR156 Pin	Type	Description
S0_RXP	50	62	Output	Receiver data differential output pair
S0_RXN	49	60	Output	Receiver data differential output pair
S0_TXP	48	58	Input	Transmitter data differential input pair
S0_TXN	46	56	Input	Transmitter data differential input pair
S1_RXP	57	72	Output	Receiver data differential output pair
S1_RXN	56	70	Output	Receiver data differential output pair
S1_TXP	55	68	Input	Transmitter data differential input pair
S1_TXN	53	66	Input	Transmitter data differential input pair

7.2 QSGMII MAC

The LAN96459F device supports a QSGMII MAC to convey four ports of network data and port speed of 10/100/1000 Mbps. See Figure 7-1.

FIGURE 7-1: QSGMII MAC INTERFACE CONNECTIONS

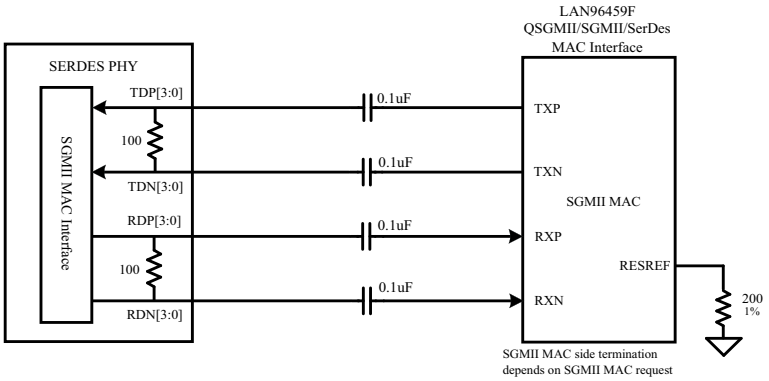


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7.3 SGMII MAC

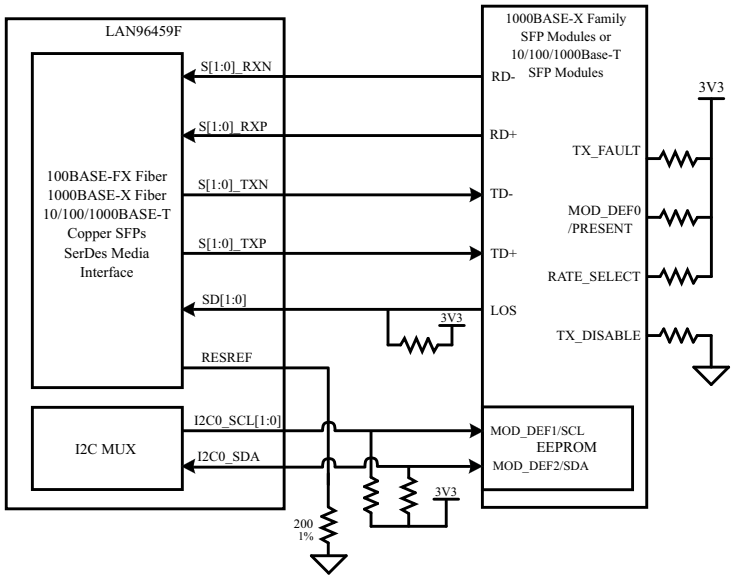
When configured to detect and switch between 10BASE-T, 100BASE-T, and 1000BASE-T data rates, the LAN96459F device can be connected to an SGMII-compatible MAC. See Figure 7-2.

FIGURE 7-2: SGMII MAC INTERFACE CONNECTIONS



7.4 SFP Ports

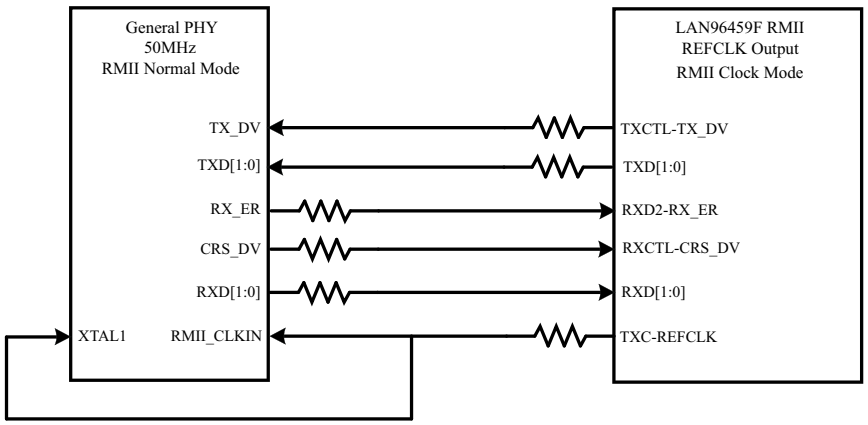
FIGURE 7-3: SERDES MEDIA INTERFACE TO 100BASE-X FAMILY FIBER SFP OR 10/100/1000BASE-T COPPER SFP REFERENCE CONNECTION



8.0 RGMII/RMII

8.1 RMII (Clock Mode)

FIGURE 8-1: CONNECTIONS BETWEEN GENERAL PHY NORMAL MODE AND LAN96459F RMII CLOCK MODE



Note 1: **RXD2-RXER** is optional in RMII mode.

2: **RMII_CLKIN** is optional on many PHYs.

8.2 RMII (Normal Mode)

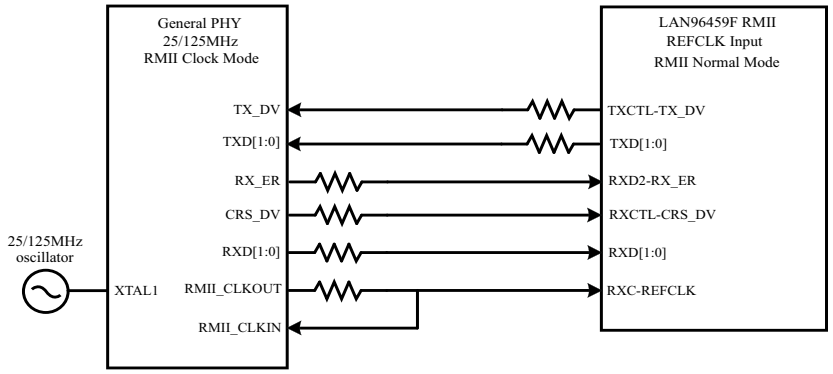
This section provides information about the RMII functionality. The device includes two RMII interfaces: RMII_0 and RMII_1. RMII is a reduced MII interface with a maximum speed of 100 Mbps.

The RMII interfaces can be used by a limited number of devices:

- RMII_0: Port Module 4 or Port Module 7
- RMII_1: Port Module 8. The RMII interfaces must be enabled and assigned to a device before they can be used.
- RGMII_0_CFG and RGMII_1_CFG: Assign RMII to a device. These registers are shared with RGMII mode configuration
- RMII_ENA: Enable RMII ports on GPIOs
- RGMII_ENA: Must be 0
- GMII_ENA: Must be 0
- REF_CLK_SEL: Select source of 50 MHz reference clock. 0 = internal, 1 = external.
- The CuPHY must use the REF_CLK output from GPIO6 and GPIO18 as REF_CLK in Internal mode.
- RMII_RX_RST=0 and RMII_TX_RST=0: Enable RMII interface by releasing Reset
- Configure MACs to match RGMII link Speed and Duplex mode

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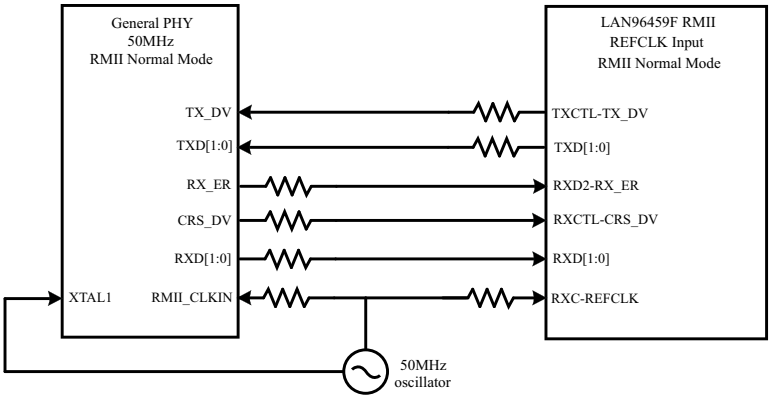
FIGURE 8-2: CONNECTIONS BETWEEN GENERAL PHY CLOCK MODE AND LAN96459F RMII NORMAL MODE



Note 1: **RXD2-RXER** is optional in RMII mode.

2: **RMII_CLKIN** is optional on many PHYs.

FIGURE 8-3: CONNECTIONS BETWEEN GENERAL PHY NORMAL MODE AND LAN96459F RMII NORMAL MODE



Note 1: **RXD2-RXER** is optional in RMII mode.

2: **RMII_CLKIN** is optional on many PHYs.

8.3 RGMII

The device includes two RGMII interfaces: RGMII_0 and RGMII_1. RGMII is a reduced GMII interface with a maximum speed of 1000 Mbps.

The RGMII interfaces can be used by a limited number of devices:

- RGMII_0: Port Module 4 or Port Module 7
- RGMII_1: Port Module 8

The RGMII interfaces must be enabled and assigned to a device before they can be used:

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- RGMII_0_CFG and RGMII_1_CFG: Assign RGMII to a device.
- RGMII_ENA: Enable RGMII ports on GPIOs.
- GMII_ENA: Must be 0

Optionally, enable the Internal Delay (ID) mode for selected clocks using these steps:

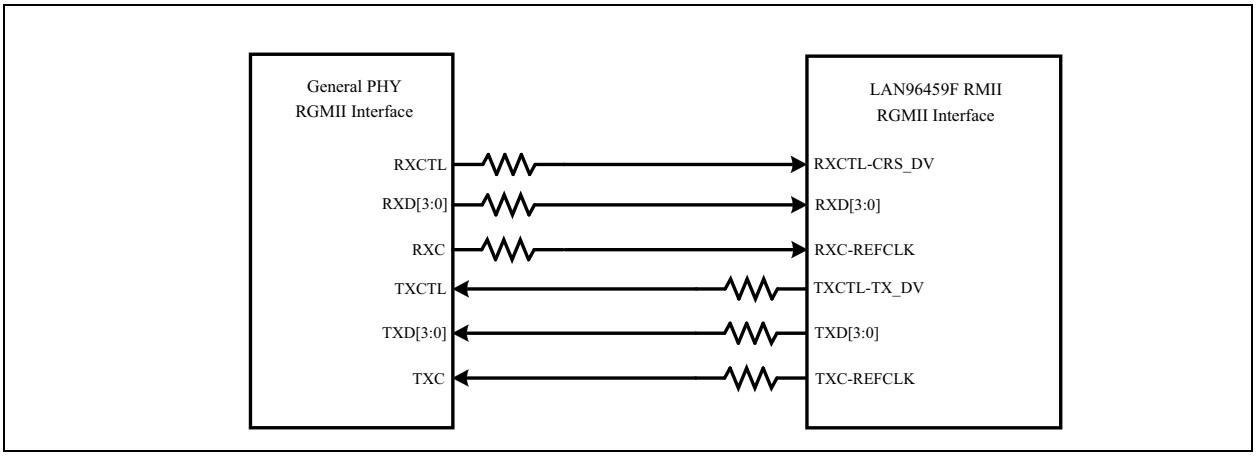
1. Bring DLL out of Reset: **DLL_RST=0**
2. Start delay tuning state machine: **DLL_EN=1**
3. Enable delayed clock output from DLL: **DELAY_EN=1**
4. Use DLL clock: **DLL_CLK_ENA=1**

- TX_CLK_CFG: Configure RGMII link speed.

  0 = Disabled, 1 = 1000 Mbps, 2 = 100 Mbps, 3 = 10 Mbps

  The MAC must be configured to use the same speed.
- RGMII_RX_RST=0 and RGMII_TX_RST=0: Enable RGMII interface by releasing Reset.
- Configure MACs to match RGMII link speed and Duplex mode.

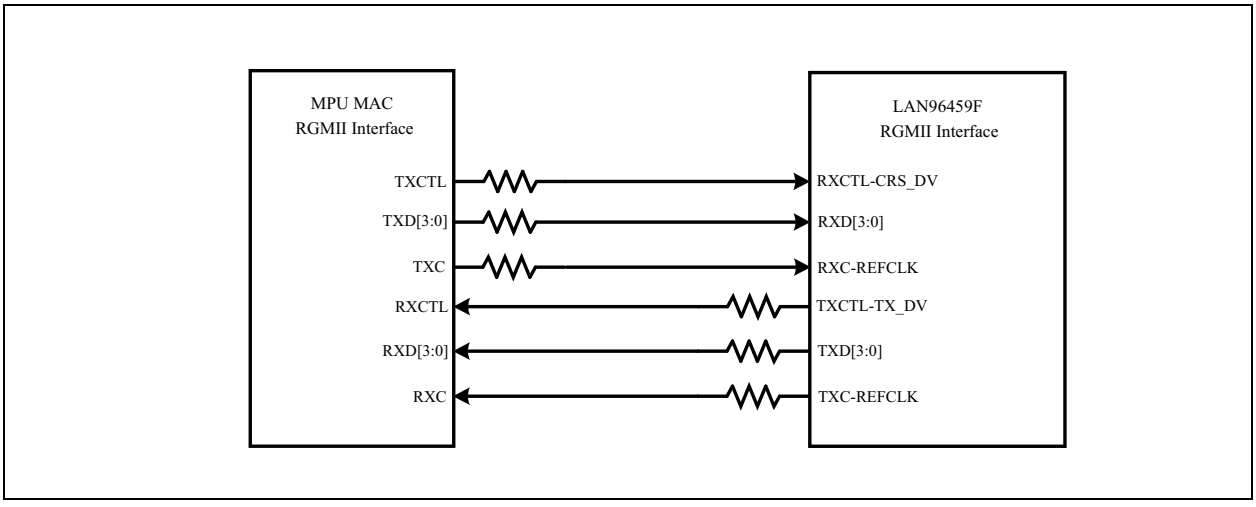
FIGURE 8-4: CONNECTIONS BETWEEN GENERAL PHY CLOCK MODE AND LAN96459F RGMII



8.4 RGMII/RMII Connection to other MAC Device

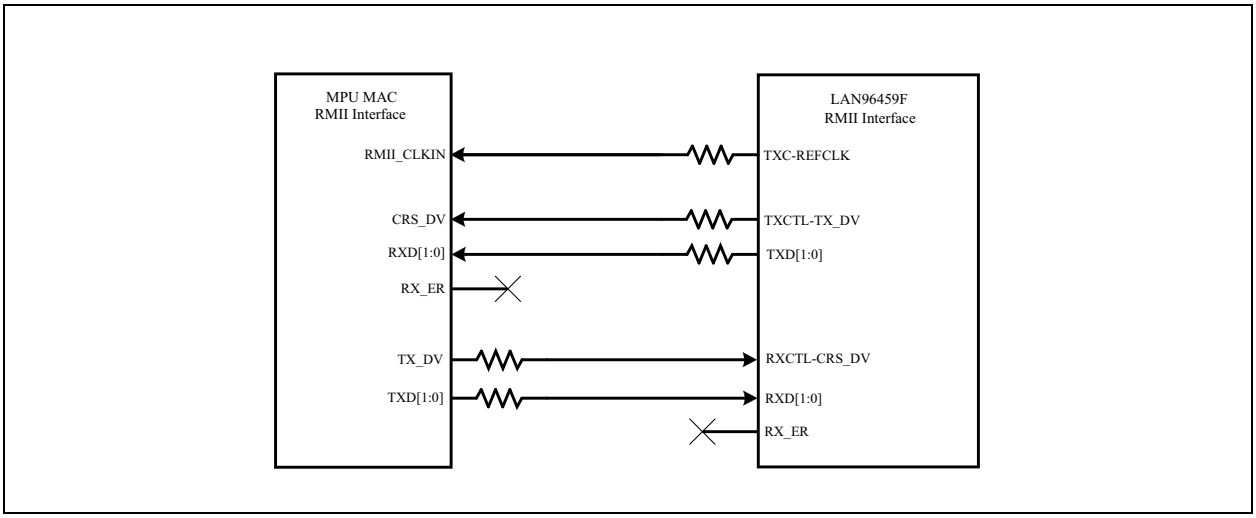
This section describes the recommended methods for connecting the LAN96459F RGMII and RMII interfaces to external MAC devices, such as CPUs, MPUs, or SoCs.

FIGURE 8-5: RGMII MAC CONNECTION



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FIGURE 8-6: RMII MAC CONNECTION



9.0 MANAGEMENT INTERFACES

9.1 I²C

TABLE 9-1: I²C CONTROLLER PIN NAMES

Name	Description	TQFP128 Pin (Alt Assignment)	VQFN-DR156 Pin (Alt Assignment)
I2C0_SDA	Two-wire serial interface data	Pin_63 - GPIO_30 (ALT2)	Pin_77-GPIO_30 (ALT2)
I2C0_SCL0	Two-wire serial interface gated clocks	Pin_62 - GPIO_31 (ALT2)	Pin_76- GPIO_31 (ALT2)
I2C0_SCL1	Two-wire serial interface gated clocks	Pin_61 - GPIO_32 (ALT2)	Pin_75- GPIO_32 (ALT2)
I2C0_SCL2	Two-wire serial interface gated clocks	Pin_95- GPIO_4 (ALT2)	Pin_112- GPIO_4 (ALT2)
I2C0_SCL3	Two-wire serial interface gated clocks	Pin_94- GPIO_5 (ALT2)	Pin_111- GPIO_5 (ALT2)
I2C0_SCL4	Two-wire serial interface gated clocks	Pin_93- GPIO_6 (ALT2)	Pin_110- GPIO_6 (ALT2)
I2C0_SCL5	Two-wire serial interface gated clocks	Pin_91- GPIO_7 (ALT2)	Pin_108- GPIO_7 (ALT2)
I2C0_SCL6	Two-wire serial interface gated clocks	Pin_90- GPIO_8 (ALT2)	Pin_107- GPIO_8 (ALT2)
I2C0_SCL7	Two-wire serial interface gated clocks	Pin_89- GPIO_9 (ALT2)	Pin_106- GPIO_9 (ALT2)
I2C0_SCL8	Two-wire serial interface gated clocks	Pin_87- GPIO_10 (ALT2)	Pin_104- GPIO_10 (ALT2)
I2C0_SCL9	Two-wire serial interface gated clocks	Pin_86- GPIO_11 (ALT2)	Pin_103- GPIO_11 (ALT2)
I2C0_SCL10	Two-wire serial interface gated clocks	Pin_85- GPIO_12 (ALT2)	Pin_102- GPIO_12 (ALT2)
I2C0_SCL11	Two-wire serial interface gated clocks	Pin_83- GPIO_13 (ALT2)	Pin_100- GPIO_13 (ALT2)
I2C0_SCL12	Two-wire serial interface gated clocks	Pin_82- GPIO_14 (ALT2)	Pin_99- GPIO_14 (ALT2)
I2C0_SCL13	Two-wire serial interface gated clocks	Pin_81- GPIO_15 (ALT2)	Pin_98- GPIO_15 (ALT2)
I2C1_SDA	Two-wire serial interface data	Pin_98- GPIO_2 (ALT3), Pin_95- GPIO_4 (ALT3)	Pin_116- EGPIO_44 (ALT2), Pin_119- GPIO_2 (ALT3), Pin_112- GPIO_4 (ALT3)
I2C1_SCL	Two-wire serial interface clock	Pin_97- GPIO_3 (ALT3), Pin_94- GPIO_5 (ALT3)	Pin_115- EGPIO_45 (ALT2), Pin_118- GPIO_3 (ALT3), Pin_111- GPIO_5 (ALT3)

TABLE 9-2: I²C TARGET PIN NAMES

Name	Description	TQFP128 Pin (Alt Assignment)	VQFN-DR156 Pin (Alt Assignment)
I2C_SLV_SDA	Two-wire serial interface data	Pin_100- GPIO_0 (ALT3)	Pin_44-EGPIO_49 (ALT3), Pin_121- GPIO_0 (ALT3)
I2C_SLV_SCL	Two-wire serial interface clock	Pin_99-GPIO_1 (ALT3)	Pin_45-EGPIO_50 (ALT3), Pin_120-GPIO_1 (ALT3)

9.2 SPI

TABLE 9-3: SPI CONTROLLER PIN NAMES

Name	Description	TQFP128 Pin (Alt Assignment)	VQFN-DR156 Pin (Alt Assignment)
SI_nCS	Active-low chip select	Pin_98- GPIO_2 (ALT1), Pin_33- GPIO_42 (ALT1)	Pin_119- GPIO_2 (ALT1), Pin_39- GPIO_42 (ALT1)

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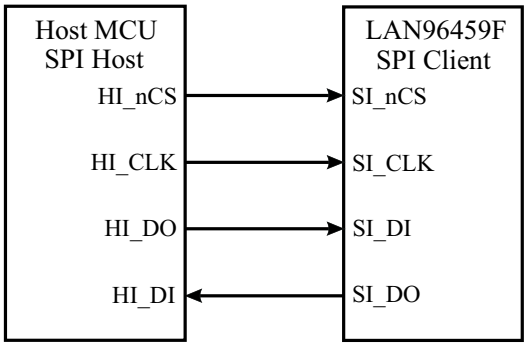
TABLE 9-3: SPI CONTROLLER PIN NAMES (CONTINUED)

Name	Description	TQFP128 Pin (Alt Assignment)	VQFN-DR156 Pin (Alt Assignment)
SI_CLK	Clock input	Pin_97- GPIO_3 (ALT1), Pin_31- GPIO_43 (ALT1)	Pin_118- GPIO_3 (ALT1), Pin_36- GPIO_43 (ALT1)
SI_DI	Data input	Pin_100- GPIO_0 (ALT1), Pin_35- GPIO_40 (ALT1)	Pin_121- GPIO_0 (ALT1), Pin_40- GPIO_40 (ALT1)
SI_DO	Data output	Pin_99- GPIO_1 (ALT1), Pin_34- GPIO_41 (ALT1)	Pin_120- GPIO_1 (ALT1), Pin_41- GPIO_41 (ALT1)

TABLE 9-4: SPI TARGET PIN NAMES

Name	Description	TQFP128 Pin (Alt Assignment)	VQFN-DR156 Pin (Alt Assignment)
SI_SLV_nCS	Active-low chip select	Pin_98- GPIO_2 (ALT2), Pin_93- GPIO_6 (ALT6)	Pin_119- GPIO_2 (ALT2), Pin_110- GPIO_6 (ALT6)
SI_SLV_CLK	Clock input	Pin_97- GPIO_3 (ALT3), Pin_91- GPIO_7 (ALT7)	Pin_118- GPIO_3 (ALT3), Pin_108- GPIO_7 (ALT7)
SI_SLV_DI	Data input	Pin_100- GPIO_0 (ALT2), Pin_95- GPIO_4 (ALT4)	Pin_121- GPIO_0 (ALT2), Pin_112- GPIO_4 (ALT4)
SI_SLV_DO	Data output	Pin_99- GPIO_1 (ALT2), Pin_94- GPIO_5 (ALT4)	Pin_120- GPIO_1 (ALT2), Pin_111- GPIO_5 (ALT4)

FIGURE 9-1: SPI CONNECTIONS BETWEEN MCU AND LAN96459F



9.3 MIIM

The LAN96459F has two MII-Management (MIIM) controllers with equal functionality. Figure 9-2 shows the layout scheme with split end-termination on MDC of 330Ω when connecting the MIIM_0 interface to multiple PHYs. Table 9-5 lists the GPIO pins.

FIGURE 9-2: CONNECTING MDC/MDIO TO MULTIPLE PHYS

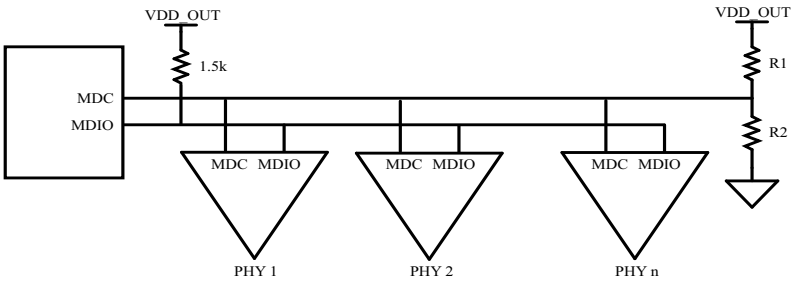


TABLE 9-5: MIIM INTERFACE ON GPIO PINS

Name	Description	TQFP128 Pin (Alt Assignment)	VQFN-DR156 Pin (Alt Assignment)
MIIM0_MDC	Data clock	Pin_65-GPIO_28 (ALT2)	Pin_79-GPIO_28 (ALT2)
MIIM0_MDIO	Data I/O	Pin_64-GPIO_29 (ALT2)	Pin_78-GPIO_29 (ALT2)
MIIM1_MDC	Data clock	Pin_86-GPIO_11 (ALT3), Pin_91-GPIO_7 (ALT5), Pin_100-GPIO_0 (ALT6)	Pin_116-EGPIO_44 (ALT1), Pin_103-GPIO_11 (ALT3), Pin_108-GPIO_7 (ALT5), Pin_121-GPIO_0 (ALT6)
MIIM1_MDIO	Data I/O	Pin_85-GPIO_12 (ALT3), Pin_90-GPIO_8 (ALT5), Pin_99-GPIO_1 (ALT6)	Pin_115-EGPIO_45 (ALT1), Pin_102-GPIO_12 (ALT3), Pin_107-GPIO_8 (ALT5), Pin_120-GPIO_1 (ALT6)
MIIM_SLV_ADDR	Slave address	Pin_100-GPIO_0 (ALT4)	Pin_46-EGPIO_48 (ALT1), Pin_121-GPIO_0 (ALT4)
MIIM_SLV_MDC	Slave data clock	Pin_99-GPIO_1 (ALT4)	Pin_44-EGPIO_49 (ALT1), Pin_120-GPIO_1 (ALT4)
MIIM_SLV_MDIO	Slave data I/O	Pin_97-GPIO_3 (ALT4)	Pin_45-EGPIO_50 (ALT1), Pin_118-GPIO_3 (ALT4)

Note: For MIIM client address:

- When 1, select the MIIM client address “11111.”
- When 0, select the MIIM client address “00000.”

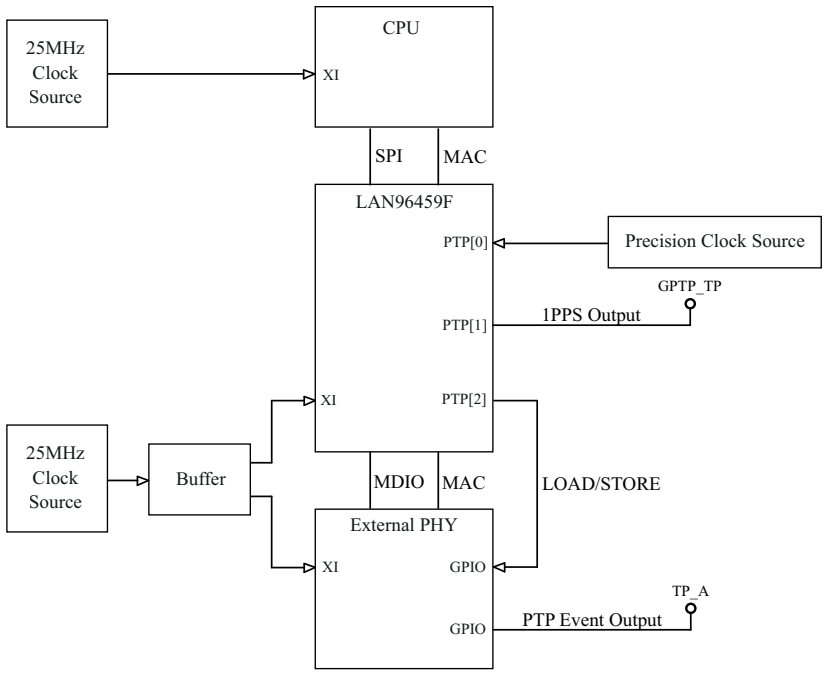
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10.0 1588 SUPPORT (PTP)

The LAN96459F device supports hardware-assisted PTP with high-accuracy time stamping logic and logic for adjusting the local time-of-day (TOD) counter to synchronize to an external timing server through GPIO signals. Time of day is maintained in a counter consisting of 48 bits of seconds, 30 bits of nanoseconds, and 8 bits of sub-nanoseconds. This counter set is called a Local Timing Clock (LTC). For further details, see the Hardware Time Stamping section of the *LAN9645xF Data Sheet*.

The LAN96459F device has four (4) GPIOs that are PTP enabled. These are detailed in Table 10-1. Figure 10-1 gives a reference for connections from these pins. This configuration supports both 1-step and 2-step modes.

FIGURE 10-1: 1-STEP OR 2-STEP PTP



Note 1: PTP[0], PTP[1], and PTP[2] may be any of the PTP GPIO pins available. However, each must be a different pin. Reference Table 10-1 for available pins.

2: Precision clock source is application dependent and, therefore, optional.

3: The 1PPS Output and PTP Event Output are used for testing and calibration.

TABLE 10-1: PTP PINS

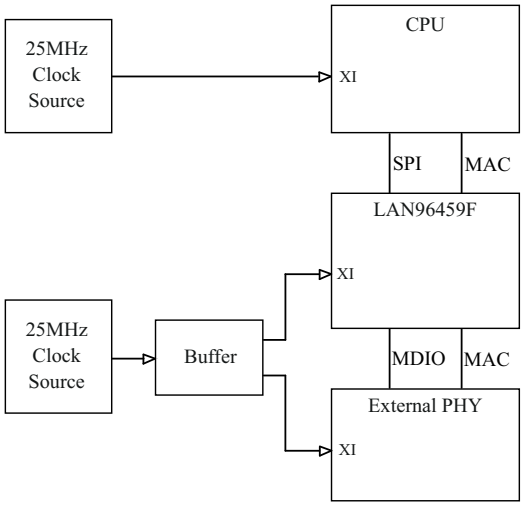
Name	TQFP128 Pin	VQFN-DR156 Pin
GPIO_12	85	102
GPIO_13	83	100
GPIO_14	82	99
GPIO_15	81	98

Note 1: GPIO_12, GPIO_13, GPIO_14, and GPIO_15 are configuration straps. See Section 4.1, "Configuration Straps".

The LAN96459F device also supports end-to-end transparent clock mode. See Figure 10-2.

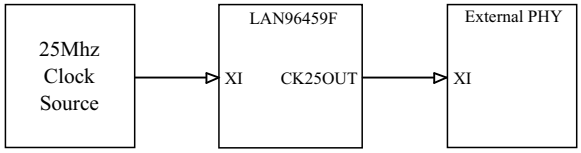
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FIGURE 10-2: TRANSPARENT CLOCK MODE



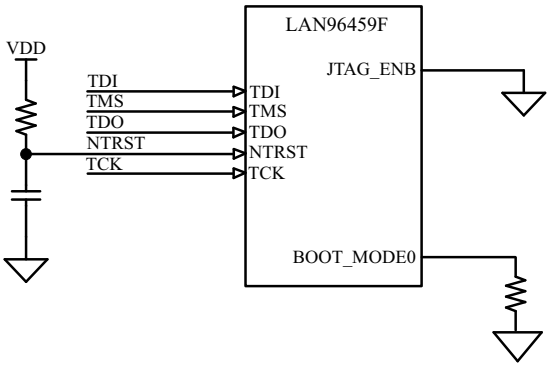
In the 156-Pin VQFN-DR package, a single crystal oscillator can be connected to the LAN96459F device. Then, the **CK25OUT** pin of the LAN96459F device can be used to provide the clock source to a PHY. See Figure 10-3.

FIGURE 10-3: 156-PIN VQFN-DR PTP EXTERNAL CRYSTAL CONNECTIONS



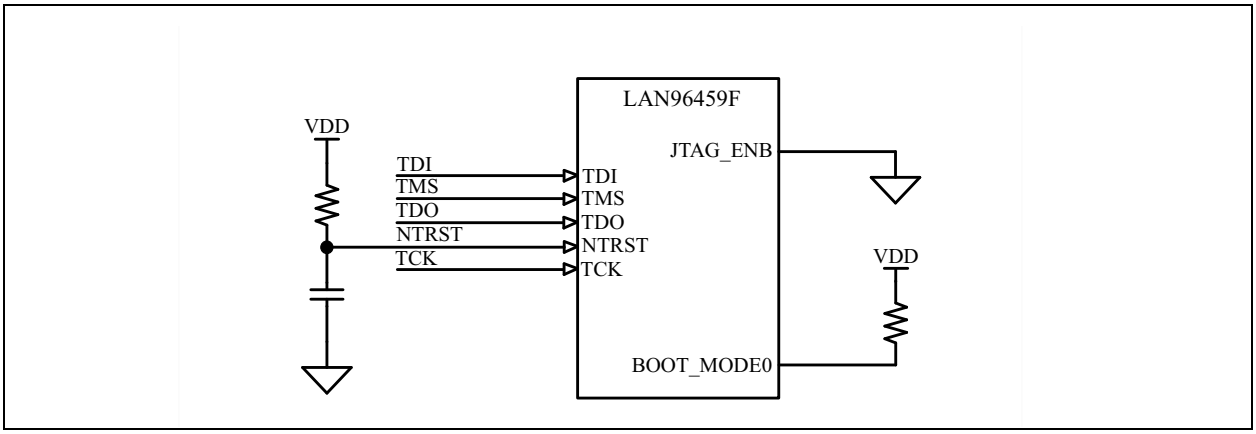
11.0 JTAG INTERFACE

FIGURE 11-1: JTAG ENABLE MODE (BOOT_MODE0 PULL-DOWN OPTION)



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FIGURE 11-2: JTAG ENABLE MODE (BOOT_MODE0 PULL-UP OPTION)



When JTAG is enabled, configuration strap **BOOT_MODE0** selects:

- Pull-down = 0: DFT TAP Controller
- Pull-up = 1: SerDes TAP Controller

FIGURE 11-3: JTAG DISABLE MODE

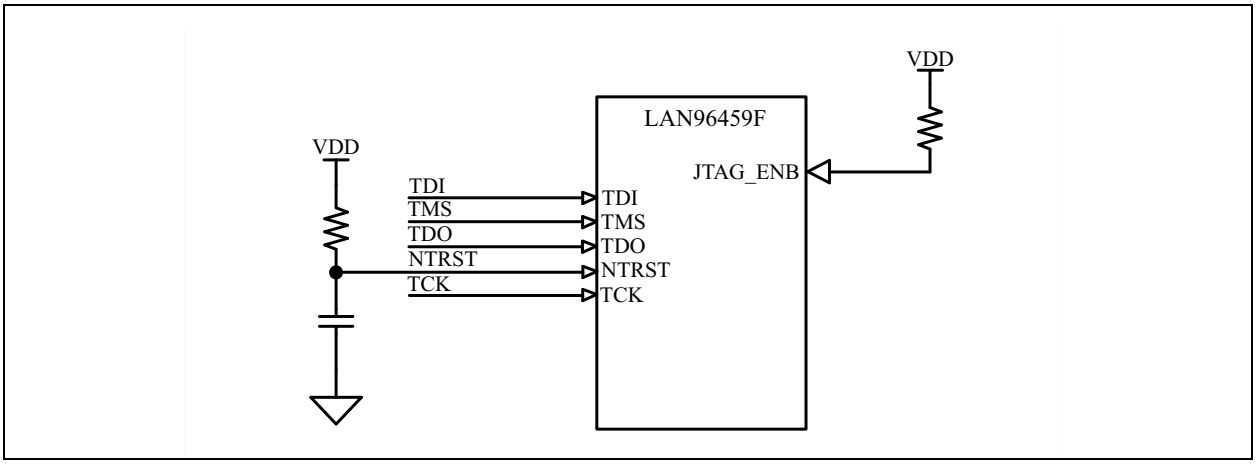


TABLE 11-1: JTAG PIN NAMES

Name	Description	TQFP128 Pin	VQFN-DR156 Pin
TDI	JTAG (IEEE 1149.1) data input	Pin_36- GPIO_39	Pin_42- GPIO_39
TCK	JTAG (IEEE 1149.1) test clock	Pin_35- GPIO_40	Pin_40- GPIO_40
TDO	JTAG (IEEE 1149.1) test data output	Pin_34- GPIO_41	Pin_41- GPIO_41
NTRST	JTAG (IEEE 1149.1) TAP controller test	Pin_33- GPIO_42	Pin_39- GPIO_42
TMS	JTAG (IEEE 1149.1) test mode select	Pin_31- GPIO_43	Pin_36- GPIO_43

Note 1: For all JTAG pins, the GPIO mode must be Manual.

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12.0 LAYOUT CONSIDERATIONS

The following standard high-speed design layout techniques must be followed:

- Minimize the number of signal-vias.
- No 90-degree bends on SerDes signals; 45 degree bends or better curved traces should be used.
- Traces must refer to a single power or ground plane.
- Do not let high-speed traces pass plane splits on an adjacent layer.
- To minimize crosstalk, the characteristic impedances of signals should be determined predominantly by the distance to the reference plane and not the distance to the neighboring traces.
- Allow as much clearance between traces as possible.
- Do not let two traces neighbor at a fixed low distance for extended lengths of trace.
- Reduce trace length as much as possible, especially for very high-speed differential signals (for example, QSGMII).

Special care must be taken for the high-speed differential pairs (QSGMII/SGMII/PCIe):

- Must be routed as 100Ω differential traces.
- Allow as much clearance as possible between differential pairs.
- The well-known rule-of-thumb of “3*W” (W = width of trace) is absolute minimum, allow for 5*W or even 10*W clearance wherever possible to minimize crosstalk.
- If guard traces or planes are used between individual pairs, use the same clearance constraints trace-to-plane as pair-to-pair (for example, 3*W, 5*W, or 10*W) to reduce the impact of the guard on the impedance of especially the trace (be that P or N) closest to the guard. Tie the guard to the inner layer ground plane frequently along the way. Untied guards are good antennas.
- Intra-pair skew (delta between lengths of P and N traces) must be as low as possible. For example, for a 5 Gbit/second trace, an intra-pair skew of 3 mm translates into 20 ps or 10% of a bit, causing the differential receiver to read the signal with excessive jitter. Nonchalance and high-speed do not match.
- 10/100/1000BASE-T signals are sensitive analog signals that will be transmitted on up to 100 meters of UTP cable.
- The analog signals must be routed from the PHY to magnetics, and from magnetics to the RJ45 connector, as four 100Ω differential pairs for each port.
- It is important to keep these signals away from noise sources. Differential (and some Common-mode) noise picked up by the signals between PHY and magnetics will end up as radiated emissions on the UTP cable. Differential, as well as Common-mode, noise picked up by the signals between magnetics and RJ45 connector will end up as radiated emissions on the UTP cable.
- It is also important to keep the two traces that make up a differential pair same length. Any difference in length will result in Differential-to-Common mode conversion of the signal, and this will also end up as radiated emissions on the UTP cable.

Low-bandwidth serial interfaces can have fast edges, too:

- Even though the effective bandwidths of these interfaces are low, it is important to avoid reflections on the multi-drop clocks (MDC, I2C_SCL, SGPIO_CLK, and SPI_CLK). Route the clock nets daisy-chained from the source to first consumer, then to the next consumer, and so on, and finally to the parallel termination network.
- The pull-ups on **MDIO** and **I2C_SDA** must be present and can be placed where convenient.

All power supplies should be routed as planes. Below are additional constraints:

- Limit via sharing (where several power pins are connected to a single via). Ideally, each power pin should be connected to its associated power plane through a dedicated via.
- Surface-mount local decoupling capacitors must be used and should be placed as close to the power supply pins as possible.
- Analog supplies must be isolated from the remaining board supplies using ferrite beads.
- Use a single ground point for the REF_REXT/REF_FILT resistor and capacitor. Simulate the completed design, and possibly selected signals during layout:
  - Simulate power integrity, for example, power drops/current densities. Even though Microchip devices are low-power, small voltage drops on, for example, a 1.1V supply quickly add up to several percent of nominal voltage. If power integrity simulation shows issues, these can possibly be solved by using wider planes, thicker planes (for example, ½ oz to 2 oz copper), or, in some cases, a remote sense point (located at the power consumer) for the DC/DC converter circuit.

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- Simulate signal integrity. SGMII/SerDes/QSGMII signals are high-speed. In some cases, they are extremely high-speed that it is much preferable to simulate on the layout file than to measure on the actual board.
- If a full simulation is not possible, simulating a single lane of a multi-lane interface is a recommended alternative as it can still provide valuable insight. Microchip supplies IBIS (or HSPICE) models for its devices' high-speed inputs/outputs.

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NOTES:

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13.0 HARDWARE CHECKLIST SUMMARY

TABLE 13-1: HARDWARE DESIGN CHECKLIST

Section	Check	Explanation	√	Notes
Section 2.0, "General Considerations"	Section 2.1, "Required References"	All necessary documents are on hand.		
	Section 2.2, "Pin Check"	The pins match the data sheet.		
Section 3.0, "Power Supplies and Grounding"	Section 3.1, "Power Supplies"	Verify that power pin connections are made based on Figure 3-1 or Figure 3-2 and that the connected power planes are decoupled based on Figure 3-3.		
	Section 3.2, "Power Pins"	Refer to Table 3-1 to ensure correct power pin usage.		
	Section 3.3, "Ground"	Refer to Table 3-2 to ensure correct ground pin usage.		
	Section 3.4, "RESET Circuit"	Refer to Figure 3-4 for Reset configuration.		
Section 4.0, "Pin Strapping"	Section 4.1, "Configuration Straps"	Refer to Table 4-1 for configuration strap utilization.		
	Section 4.2, "LED"	Refer to Table 4-2 for LED utilization. Refer to Table 4-3 and Figure 4-1 to verify GPIO LED and strap overlaps and configuration.		
Section 5.0, "Reference Clocks"	Section 5.1, "Layer 1 Timing"	If used, verify SerDes recovered clock configuration registers. Refer to Table 5-1, Table 5-2, Table 5-3, and Table 5-4 to ensure register settings match the implemented output frequency and Ethernet standard.		
	Section 5.2, "Clocking"	If a 25 MHz crystal is used, refer to Figure 5-1 to ensure correct clock usage.		
		If an external 25 MHz oscillator is used, refer to Figure 5-2 to ensure correct clock usage.		
		If the 125 MHz reference clock input is used, refer to Figure 5-3 to ensure correct clock usage.		
	Section 5.3, "CK25OUT"	If the 156-Pin VQFN-DR package is used, verify that **CK25OUT** is utilized or not connected.		
Section 6.0, "Port Configuration"	Section 6.1, "Logical Ports and Ethernet Interfaces"	Refer to Figure 6-1, Table 6-1, and Table 6-2 for configuring port multiplexing.		
	Section 6.2, "CuPHY Interfaces"	Refer to Table 6-3, Figure 6-2, and Figure 6-3 to ensure correct port pin usage and configuration.		
Section 7.0, "SerDes Interface"	Section 7.1, "Serial Data Pins"	Refer to Table 7-1 to ensure correct serial data pin usage.		
	Section 7.2, "QSGMII MAC"	Refer to Figure 7-1 for QSGMII MAC interface to connect to external QSGMII MACs in the design.		
	Section 7.3, "SGMII MAC"	Refer to Figure 7-2 for SGMII MAC interface to connect to external SGMII MACs in the design.		
	Section 7.4, "SFP Ports"	Refer to Figure 7-3 for the interface with 1000BASE-X family fiber SFP OR 10/100/1000BASE-T Copper SFP.		

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TABLE 13-1: HARDWARE DESIGN CHECKLIST (CONTINUED)

Section	Check	Explanation	√	Notes
Section 8.0, "RGMII/RMII"	Section 8.1, "RMII (Clock Mode)"	Refer to Figure 8-1 to ensure correct RMII Clock mode configuration.		
	Section 8.2, "RMII (Normal Mode)"	Refer to Figure 8-2 to ensure correct RMII Normal-to-Clock mode configuration.		
		Refer to Figure 8-3 to ensure correct RMII Normal-to-Normal mode configuration.		
	Section 8.3, "RGMII"	Refer to Figure 8-4 to ensure correct RGMII Clock mode configuration.		
	Section 8.4, "RGMII/RMII Connection to other MAC Device"	Refer to Figure 8-5 and Figure 8-5 to ensure correct RGMII and RMII MAC connections, respectively.		
		Refer to Figure 8-5 to ensure correct RGMII Clock mode configuration.		
Section 9.0, "Management Interfaces"	Section 9.1, "I^2C"	Refer to Table 9-1 and Table 9-2 to ensure correct I^2C pin usage.		
	Section 9.2, "SPI"	Refer to Figure 9-1 as well as Table 9-3 and Table 9-4 to ensure correct SPI configuration and pin usage.		
	Section 9.3, "MIIM"	Refer to Figure 9-2 and Table 9-5 to ensure correct MIIM configuration and pin usage.		
Section 10.0, "1588 Support (PTP)"		Refer to Figure 10-1, Figure 10-3, and Table 10-1 to ensure correct PTP configuration and pin usage.		
		If the end-to-end transparent clock is used, refer to Figure 10-2 to ensure correct configuration.		
Section 11.0, "JTAG Interface"		Refer to Figure 11-1, Figure 11-2, Figure 11-3, and Table 11-1 to ensure correct JTAG configuration and pin usage.		
Section 12.0, "Layout Considerations"		Refer to this section for standard high-speed design layout techniques.		

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APPENDIX A: REVISION HISTORY

TABLE A-1: REVISION HISTORY

Revision Level & Date	Section/Figure/Entry	Correction
DS00006273B (02-27-26)	Figure 3-4	Changed VDD to VDD_IO_A and added Note 3.
	Figure 11-1	Added a new figure.
	Figure 11-2 and Figure 11-3	Revised the figures and notes.
	Table 11-1	Updated table.
	Section 8.4 “RGMII/RMII Connection to other MAC Device”	Added a new section.
DS00006273A (11-07-25)	Initial release	

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